

6. Matchings

What students need to learn:

Use of bipartite graphs for modelling matchings. Complete matchings and maximal matchings.

Students will be required to use the maximum matching algorithm to improve a matching by finding alternating paths. No consideration of assignment is required.

Algorithm for obtaining a maximum matching.

1. Algorithms

In a list containing N items the 'middle' item has position $\lceil \frac{1}{2}(N + 1) \rceil$ if N is odd $\lceil \frac{1}{2}(N + 2) \rceil$ if N is even, so that if $N = 9$, the middle item is the 5th and if $N = 6$ it is the 4th.

2. Algorithms on graphs

A **graph** G consists of points (**vertices** or **nodes**) which are connected by lines (**edges** or **arcs**).

A **subgraph** of G is a graph, each of whose vertices belongs to G and each of whose edges belongs to G .

If a graph has a number associated with each edge (usually called its **weight**) then the graph is called a **weighted graph** or **network**.

The **degree** or **valency** of a vertex is the number of edges incident to it. A vertex is **odd** (**even**) if it has **odd** (**even**) degree.

A **path** is a finite sequence of edges, such that the end vertex of one edge in the sequence is the start vertex of the next, and in which no vertex appears more than once.

A **cycle** (**circuit**) is a closed path, i.e. the end vertex of the last edge is the start vertex of the first edge.

Two vertices are **connected** if there is a path between them. A graph is **connected** if all its vertices are connected.

If the edges of a graph have a direction associated with them they are known as **directed edges** and the graph is known as a **digraph**.

A **tree** is a connected graph with no cycles.

A **spanning tree** of a graph G is a subgraph which includes all the vertices of G and is also a tree.

A **minimum spanning tree** (MST) is a spanning tree such that the total length of its arcs is as small as possible. (MST is sometimes called a **minimum connector**.)

A graph in which each of the n vertices is connected to every other vertex is called a **complete graph**.

4. Critical path analysis

The **total float** $F(i, j)$ of activity (i, j) is defined to be $F(i, j) = l_j - e_i - \text{duration}(i, j)$, where e_i is the earliest time for event i and l_j is the latest time for event j .

6. Matchings

A **bipartite graph** consists of two sets of vertices X and Y . The edges only join vertices in X to vertices in Y , not vertices within a set. (If there are r vertices in X and s vertices in Y then this graph is $K_{r,s}$.)

A **matching** is the pairing of some or all of the elements of one set, X , with elements of a second set, Y . If every member of X is paired with a member of Y the matching is said to be a **complete matching**.